Problem 1 ... my hand is slipping off

3 point

According to information from Wikipedia, the longest escalators in Prague are at Náměstí Míru They travel a distance $l = 87.1\,\mathrm{m}$ *in* $t = 140\,\mathrm{s}$. *It is also said that the handrails on the escalator move* $\delta v = 2.0\,\%$ *faster than the stairs in order to ensure alertness. How much would your arm move relative to the rest of your body (if it was long enough) during the entire escalator ride if you stood on the same stair while holding one spot the whole time?*

Karel was wondering about escalators again

Problem 2 ... comparison of slings

3 point

Lego got bored of bows, so he went to buy a sling. We approximate it as a spring with constan stiffness and a zero rest length. Lego can stretch the sling with force $F = 31.4\,\mathrm{N}$*, and he wants i to store as much energy as possible while it is stretched. There were two slings at the shop, wit stiffnesses* $k_1 = 159\,\mathrm{N/m}$ *and* $k_2 = 265\,\mathrm{N/m}$*. What will be the difference between the potentia energies of those slings stretched with force* F*? Give a positive result if the sling with* k_1 *ha more energy and a negative one if the opposite is true.*

Lego got bored of bows

Problem 3 ... mountain biking

3 point

Matěj is riding on a mountain bike trail, passing oncoming cyclists. At the top of the hil exhausted riders are moving at an average speed of $5\,\mathrm{km{\cdot}h^{-1}}$ *in both directions and Matě meets them here at an average rate of* 0.02 *per second. On the other hand, at the lowest poin of the trail, the cyclists are revved up (from both directions) and moving at an average speed o* $0\,\mathrm{km{\cdot}h^{-1}}$*. How many cyclists does Matěj meet here on average? The bike trail has no turnoffs and Matěj always moves at the speed of an average cyclist.* *Matěj mountain biked*

Problem 4 ... hunter, shoot yourself

3 point

What is the largest radius a planet can have for a person to be able to shoot oneself on it Hunter shoots just above its surface, parallel to it, and the planet will have no atmosphere The velocity of a flying bullet is $v = 380\,\mathrm{m\cdot s^{-1}}$*, and the density of the planet is* $\rho = 3\,200\,\mathrm{kg\cdot m^{-3}}$ *Furthermore, assume the planet to be spherical and homogeneous.*

Karel wanted planetary shooting

Problem 5 ... falling icicles

3 point

A small icicle broke loose from the edge of the roof and started its free fall along the vertica wall of the house. The icicle flew past a window of height $h = 1.50\,\mathrm{m}$ during a time $t = 0.10\,\mathrm{s}$ What is the distance between the edge of the roof and the top of the window?

Kuba was once almost hit by an icicle

Problem 6 ... worn tire

3 point

The tires on our new car have a diameter of $D = 634.5\,\mathrm{mm}$ *and a tread depth of* $d_0 = 8.0\,\mathrm{mm}$. *The technician set the car's speedometer to measure accurately for this original wheel size. However, we drive frequently, so after a while, the tire tread came down to the minimum depth for summer use,* $d_{\min} = 1.6\,\mathrm{mm}$. *How does the actual speed now differ from the speedometer reading? Give the answer as the ratio of the speed on the speedometer to the actual speed.*

Karel was refueling

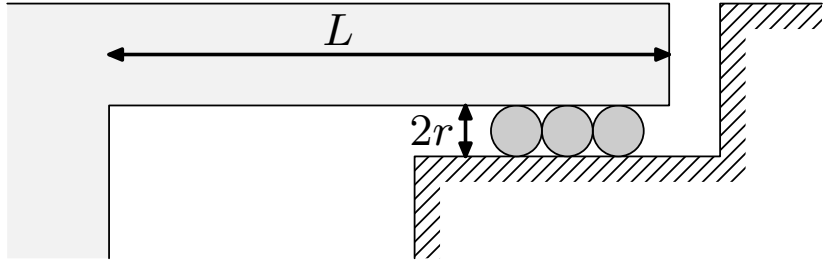
Problem 7 ... Drop Da Bomb

3 point

At the end of The Simpsons Movie, Homer and Bart are riding a motorcycle on the insid of a glass dome. Assume the dome has a radius of 1.00 km*. What minimum speed does th motorcycle have to travel at to ensure that Homer and Bart do not fall off even at the highes point of the dome? Jindra was testing his new motorcycle at The Ondřejov Observatory*

Problem 8 ... moving bridge

3 point



A road bridge of length $L = 8\,\mathrm{m}$ is firmly anchored on one side. The other side is placed on sturdy cylinders of radius $r = 6\,\mathrm{cm}$. The cylinders lay on a rigid flat surface on which they can move freely in a horizontal direction. By what angle will the cylinders rotate between winter, when the temperature is $t_z = -10\,^\circ\mathrm{C}$, and summer, when the temperature is $t_l = 30\,^\circ\mathrm{C}$? The coefficient of linear thermal expansion of the bridge is $\alpha = 1.0 \cdot 10^{-5}\ \mathrm{K}^{-1}$. Note that the cylinders do not slip and that there is a gap between the road and the moving end of the bridge.

Jarda can feel the ground moving beneath his feet

Problem 9 ... two rectangular cuboids on a hill

4 point

Let us have "a hill" made of two inclined planes with inclinations α_1 and α_2 with respect t he horizontal plane. We lay a rectangular cuboid with the mass $m_1 = 10\,\mathrm{kg}$ on the first plan with inclination $\alpha_1 = 25°$. On the second plane with inclination $\alpha_2 = 35°$, we place a differen ectangular cuboid with the mass $m_2 = 15\,\mathrm{kg}$. Then we connect the cuboids with a massless rop hat is led through a massless pulley at the top of the hill. The rope is always stretched paralle o the given plane, the coefficient of dynamic friction between the cuboids and inclined plane s $f = 0.15$, and the coefficient of static friction is small enough for cuboids to start moving What is the acceleration of the cuboid m_1 if both cuboids are at rest at the beginning? Th positive sign will indicate acceleration downhill, and a negative sign will indicate acceleratio uphill.

Lego was hiking

Problem 10 ... energy-saving electric kettle

4 point

When prices rise, you must save. What percentage of our finances will we save if we heat less wa er for tea and more economically at the same time? The original method was to heat $V_1 = 1.00$ *f water from the original temperature* $t_0 = 18.5\,^{\circ}\mathrm{C}$ *(both the kettle and water temperature o the boiling point* $t_\mathrm{v} = 98.5\,^{\circ}\mathrm{C}$*. We use a kettle with heat capacity* $C = 442\,\mathrm{J{\cdot}K^{-1}}$*. The new method will be different in a smaller volume of heated water* $V_2 = 0.800\,\mathrm{l}$ *and in a lower reache emperature of* $t_2 = 83\,^{\circ}\mathrm{C}$*. Regardless of the current price development, consider the price o lectricity for consumers as constant* $p = 4.45\,\mathrm{Kč{\cdot}kWh^{-1}}$*. Assume that the efficiency of th ettle is constant* $95\,\%$ *and that the temperature is the same in the whole volume at any give moment.*

Karel was looking at the rising inflation

Problem 11 ... induced voltage reloaded

3 point

Let us have an area of a homogenous magnetic field. The area has a rectangular cross-sectio with sides $a = 3.0\,\mathrm{m}$, $b = 2.0\,\mathrm{m}$, and the magnetic field vector $B = 1.0 \cdot 10^{-3}\,\mathrm{T}$ is perpendicula to it. We place a sufficiently long wire parallel to the diagonal of the rectangle so that it a ies entirely outside of the area with the magnetic field. We connect the ends of the wire t a voltmeter.

Now we start moving the wire in a rectilinear motion at a speed of $v = 0.20\,\mathrm{m{\cdot}s^{-1}}$ perpen dicularly to the direction of the magnetic field vector and the diagonal to which it is paralle What is the peak magnitude of voltage displayed by the voltmeter during the movement of th wire?

Vojta was proofreading

Problem 12 ... Archimedeskugeln

5 point

Consider two identical containers of mass $M_\mathrm{n} = 340.0\,\mathrm{g}$ *with the identical amount of wate of mass* $M_\mathrm{w} = 150.0\,\mathrm{g}$. *We place each of the containers on a scale separately. Into the firs container, we immerse a steel ball with density* $\rho_\mathrm{o} = 7\,850\,\mathrm{kg{\cdot}m^{-3}}$, *which is suspended fro an external frame that does not stand on the scale. The ball is fully immersed and does no touch the bottom. In the second container, we attach a ball made of polystyrene foam wit density* $\rho_\mathrm{p} = 27.0\,\mathrm{kg{\cdot}m^{-3}}$ *and radius* $r_\mathrm{p} = 1.5\,\mathrm{cm}$ *to the bottom with a thin string. The ball i fully submerged, not breaking the surface. What must be the radius of the steel ball so the tw scales will display the same weight?*

Jindra has been learning German to hide the balls in the titl

Problem 13 ... neglecting the Earth's movement

4 point

Have you ever considered that when we calculate the free fall of a body on Earth, we neglec
the Earth's motion towards the body? This calculation corresponds to an Earth with infinit
mass. However, is it possible to measure the Earth's motion, at least in theory? According t

oday's physics, we generally cannot measure distances smaller than Planck's length, whic
s $l_{\mathrm{P}} = \sqrt{\hbar G/c^3} \doteq 1.616 \cdot 10^{-35}\,\mathrm{m}$. *What is the minimum mass that a body needs to have fo*
he Earth to move from its initial position by at least $n = 10^4$ *Planck's lengths from the poin*
vhen this body is released until the point when it impacts the Earth? The body is released a
he height $h = 1.00\,\mathrm{m}$ *above the Earth's surface. Neglect the Earth's rotation, atmosphere, an*
ny other motion of the Earth. *Karel was thinking about approximations and Planck units*

Problem 14 ... hot light bulb

4 point

Danka has a lamp that has an old light bulb with an input power of $P = 60\,\mathrm{W}$ *on her desk At a distance* $h = 40\,\mathrm{cm}$ *below the bulb, there is a computer mouse with an elliptical cross ection. The semi-major axis of the ellipse is* $a = 60\,\mathrm{mm}$ *and the semi-minor* $b = 30\,\mathrm{mm}$ *When the light bulb was turned on, the mouse had a room temperature* $T_1 = 23.0\,^{\circ}\mathrm{C}$*. What will be the temperature of the mouse after* 90 *minutes of illumination? Assume that the bul adiates isotropically into the entire space,* $\eta = 83\%$ *of energy hitting the mouses cross-sectio s converted into heat, while mouse is not losing heat itself. The heat capacity of the mous s* $C = 200\,\mathrm{J{\cdot}K^{-1}}$*.*

Danka's mouse is warming up from the lamp

Problem 15 ... bead with spring

4 point

A bead of mass $m = 358\,\mathrm{g}$ is strung on a straight wire (so the bead can only move in on direction). We attach to the bead an ideal spring with zero rest length and spring constant k = = $979\,\mathrm{N{\cdot}m^{-1}}$. We fix the other end of this spring at a distance $l = 323\,\mathrm{mm}$ from the line o which the wire lies. What will be the period of the small oscillations of the bead around it equilibrium position? Lego hasn't set a problem on small oscillation for a long time

Problem 16 ... ballad of a sinful soul

5 point

Jarda's FYKOS soul will once be relieved of its physical burden and head to heaven to mee the FYKOS bird. However, weighed down by sins, it will slip into hell, where it will be enclose in a cauldron of $V = 666\,\mathrm{cm}^3$. *It will remain preserved there at drastic conditions of* $T = 666\,^\circ$ *and* $p = 666 \cdot 10^5\,\mathrm{Pa}$, *while it will try to get rid of the sin molecules to rise again. The mola mass of sin is* $666\,\mathrm{g{\cdot}mol^{-1}}$. *How many sins will Jarda accumulate during his time at FYKOS The mass of a pure soul is* $21\,\mathrm{g}$, *and its density at normal conditions is* $0.70\,\mathrm{kg{\cdot}m^{-3}}$. *Conside the soul and the sins to be an ideal gas.* *Jarda's proposition was not politically correc*

Problem 17 ... is that what you meant, Mr. Planck? 5 point

Determine the Planck's surface tension.

Hint: Planck units are those formed by combining three fundamental physical constants the gravitational constant G, the reduced Planck constant $\hbar$ and the speed of light c. Just for the record, the standard model of quantum physics fails on Planck scales in the microworld, and the effects of quantum gravity (of which we know nothing yet) must be taken into account.

Jindra came up with a problem that has a shorter statement (without a hint) than the title

Problem 18 ... red-hot resistance wire

5 point

Consider a wide resistive wire which at temperature $T_0 = 20\,^\circ\mathrm{C}$ *has the length* $l_0 = 10\,\mathrm{m}$ *an*
esistance $R_0 = 1.23\,\Omega$. *As the current passes through it, it heats up to* $T = 100\,^\circ\mathrm{C}$, *changin*
ts dimensions and specific resistance. Our wire has a coefficient of linear thermal expansio
$\alpha_l = 2.43 \cdot 10^{-3}\,\mathrm{K}^{-1}$ *and its temperature coefficient of specific electrical resistance is* $\alpha_R =$
$= 3.92 \cdot 10^{-3}\,\mathrm{K}^{-1}$. *What will be the resistance of the wire at temperature* T *if it is connecte*

o a source such that its dimensions can change with temperature?

Karel was learning to teach and thought about material constants

Problem 19 ... golf on the hill

5 point

A golfer was confident about hitting the ball across the whole course. However, his hands wer shaking during the broadcast, and his drive was not successful at all. He underhanded the ba so that it came out at an angle $\alpha = 80^\circ$ *to the horizontal plane at speed* $v = 8.0\,\mathrm{m{\cdot}s^{-1}}$*. Th launch was made in the direction down the hill, which has a deviation from the horizonta direction of* $\beta = 10^\circ$*. The disappointed golfer turned back and did not look to see how far fro the point of drive the ball landed. Therefore, calculate this figure.*

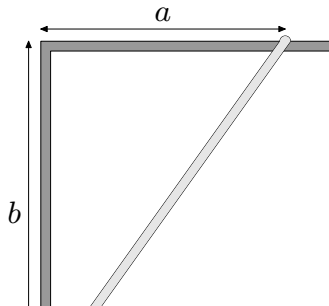
That day, Kuba lost another ball..

Problem 20 ... rubberband around the corner

4 point

We put a massless rubber band around the corner of the L-shaped rigid body. What is the smallest coefficient of friction between the rubber band and the solid that will prevent it from contracting it? The rubber band bends over the edges of the body at points $a = 5.0\,\mathrm{cm}$ *and* $b = 7.0\,\mathrm{cm}$ *from the corner.*

They're gonna put Jarda on a rack.



Problem 21 ... level sensor

6 point

Consider a vessel with a non-conductive liquid of relative permittivity $\varepsilon_r = 1.80$, in which tw vertical partially immersed parallel plates form a plate capacitor. The capacitor is connecte to a coil so that together they are part of the LC oscillator circuit. How many times doe the natural frequency of this oscillator increase when the plates of the capacitor are quarterl submerged, compared to the case when the vessel is empty? I bet that, just lik Vašek, you have already thought about a way of measuring the level in a non-mechanical way

Problem 22 ... mountain biking reloaded

4 point

Matěj is riding on a mountain bike trail, passing oncoming cyclists. At the top of the hil exhausted cyclists are moving at an average speed of $5\,\mathrm{km\cdot h^{-1}}$ *in both directions, and Matě meets there* 0.02 *oncoming bikes per second on average. In contrast, at the lowest point o the bike trail, cyclists are revved up (from both directions) and ride at an average spee of* $50\,\mathrm{km\cdot h^{-1}}$*. How many oncoming bikes do Matěj encounter here on average? The bike tra has no turnoffs, and Matěj always moves at a speed of* $10\,\mathrm{km\cdot h^{-1}}$*.*

Matěj mountain biked while reloading

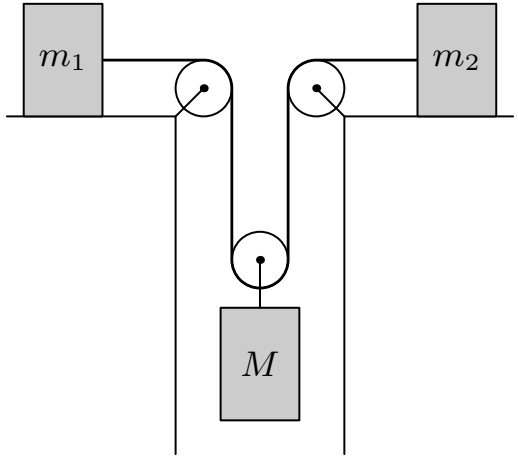
Problem 23 ... a pit with pulleys

5 point

We have a pit in which a block with mass $M = 84.6\,\mathrm{kg}$ is hanging from a movable pulley. On the horizontal surface to the left of the pit, there is a block with mass $m_1 = 26.4\,\mathrm{kg}$, and to the right of the pit, there is another block with mass $m_2 = 33.8\,\mathrm{kg}$. To each of these is tied one end of a rope, with a movable pulley hanging in the pit. There are, of course, two pulleys on the edges of the pit so that the rope runs only horizontally and vertically.

Assume that everything moves without friction; the rope and pulleys are immaterial. What is the magnitude of acceleration the block in the pit will move with?

Lego felt like he does not create enough problems with pulleys.



Problem 24 ... effective trapezoid

5 point

Consider an electrical source that provides a trapezoidal voltage such that for the first thir of the period, the voltage increases linearly from $0.00\,\mathrm{V}$ *to* $5.00\,\mathrm{V}$*, then remains constant a* $.00\,\mathrm{V}$ *for a third of the period, and then decreases linearly back to* $0.00\,\mathrm{V}$ *in the last thir of the period. We will connect the source to a resistor. What constant-voltage source coul ve replace the original source with to have the same average power on the component (i.e letermine the effective voltage of the source)?*

Karel was varying the problem on effective values

Problem 25 ... race against time

4 point

You are sitting in a car that is traveling at a constant velocity of $80\,\mathrm{km{\cdot}h^{-1}}$*, and you ente*
a $5\,\mathrm{km}$ *long tunnel with no cellular signal. At the same time, you are watching a stage o*
the Tour de France on your cell phone with $3.5\,\mathrm{km}$ *to go. The cyclists are going at* $40\,\mathrm{km{\cdot}h^{-}}$
and just as your signal drops out, they begin to descend, accelerating steadily until the finis
line. What is their largest possible acceleration so that you can see the last $30\,\mathrm{s}$ *of the stag*
when you exit the tunnel and get the signal again? *Dávid was watching cycling in the ca*

Problem 26 ... escape across the frozen lake

4 point

A migrant tries to flee the country in winter, and his path leads across a frozen lake. The borde guards are on his tail, so he must plan his route as efficiently as possible. There is a direc national border that runs across the lake, beyond which he will be safe. There are thick reed ll around the shore, but suddenly he sees a free area of ice. He runs onto the frozen surfac t a speed of $u = 4.3\,\mathrm{m{\cdot}s^{-1}}$ *in a direction parallel to the border from which he is* $d = 38\,\mathrm{m}$ *way. How long will it take him, at the minimum, to reach safety if the coefficient of frictio between the ice and his boots is* $f = 0.05$ *and no other obstacles stand in his way?*

Jarda thinks these are the real problems in Eastern Europe right nou

Problem 27 ... heating the helium

5 point

An unknown volume V_0 of helium gas is stored at atmospheric pressure and at a temperatur $T_0 = 297\,\mathrm{K}$. How large is this volume if, by supplying heat $Q = 42\,\mathrm{J}$ at constant pressure the temperature of the gas is increased by $\Delta T = 2.5\,\mathrm{K}$?

Karel was browsing the textbooks and hadn't seen this one

Problem 28 ... tropical climate

5 point

Viktor wants his house plants to thrive, so he decided to increase the humidity in the room Before turning on the humidifier, the humidity in the room was 30 %. *The humidifier ca evaporate* 5 ml *of water every minute. At what percentage would the humidity in the roo stabilize if Viktor opened the window and every minute,* 3 % *of air was replaced in the room Assume that the outside atmosphere has the same humidity as the air inside the room befor the device was turned on. Viktor's room has dimensions* $4 \times 5 \times 2.5$ m, *and the air temperatur is* 25 °C.

Viktor wanted to experience what it was like to live in the rainfores

Problem 29 ... The Pit and the Pendulum

6 point

n a cylindrical pit that is 30 cm *wide, a small ball is suspended on a* 50 cm *long massless string The point of suspension lies on the symmetry axis of the pit. We give the ball a horizonta velocity v such that it bounces perfectly elastically off the vertical walls twice before returnin o its original position. It will not pass through any point in space twice during this proces nd will have the same velocity vector at the end as at the beginning. Determine the magnitud f the velocity v.*

Jarda happened to be paying attention in literature class

Problem 30 ... magneto-resistor

5 point

Let us assemble an electric circuit. We have a long straight wire of resistance $R = 57\,\mathrm{m}\Omega$ *n parallel to the wire, we connect a magneto-resistor, i.e., a component that changes its re istance depending on the external magnetic field. The distance between the magneto-resisto nd the wire is* $d = 3.5\,\mathrm{cm}$*, and its resistance at zero magnetic field is* $r_0 = 85\,\Omega$*. In paralle o the wire, we connect a power supply with voltage* $U = 12\,\mathrm{V}$*. A current* $I = 140\,\mathrm{mA}$ *flow hrough the magneto-resistor after it has stabilized. Assume that its resistance varies linearl with the magnetic field* B *over a given range. Determine the coefficient of proportionality.*

Jarda got lost in a physicists' warehouse

Problem 31 ... wheel against the direction of travel

5 point

What fraction of the total travel time does the velocity (component parallel to the ground of one fixed point of the outermost edge of the railway wheel (the flange) have the opposit direction as the train itself? Consider that the wheel overhangs the track by 35 mm *and it diameter with the flange is* 1 400 mm.

Karel was thinking about trains

Problem 32 ... too high frequency

6 point

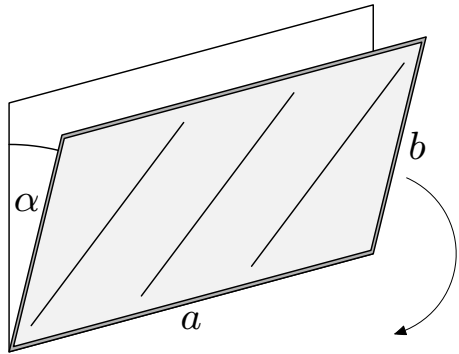
Let's have a resistor with resistance $R = 11\,\mathrm{k\Omega}$ and a capacitor with capacitance $C = 2\,\mu$ connected in series to the AC voltage source. For what value of frequency in the circuit wi the voltage amplitude across the capacitor drop to one tenth of the maximum achievable value

Jarda doesn't like quick changes

Problem 33 ... attention, the window is falling

7 point

The rectangular window is open on its upper side so that its plane makes an angle of 10° *with the vertical direction. However, due to strong wind, the latch holding it in place is released, and the window begins to rotate about its lower horizontal axis before hitting the wall below. At what angular velocity does it hit the wall? The glass alone has an area density* $\rho = 15\,\mathrm{kg\cdot m^{-2}}$*, dimensions* $a = 130\,\mathrm{cm}$ *by* $b = 60\,\mathrm{cm}$*, and is embedded in a frame of mass* $m_\mathrm{r} = 4\,\mathrm{kg}$ *(do not consider its dimensions). Consider the whole window to be planar and homogenous in parts.*



It unfortunate. Jarda has to find a glazier, but at least he has a problem for Physics Brawl Online

Problem 34 ... blocks on top of each other on a plane

5 point

Two blocks of masses 600 g *and* 700 g *are placed on top of each other on an inclined plane whic*
s deviated from the horizontal direction by 30°*. The heavier of them is on the bottom, and th*
nterface between them is perfectly smooth. The coefficient of static friction between the lowe
block and the plane is 0.3*, while the dynamic friction coefficient is* 0.2*. How large is the angl*

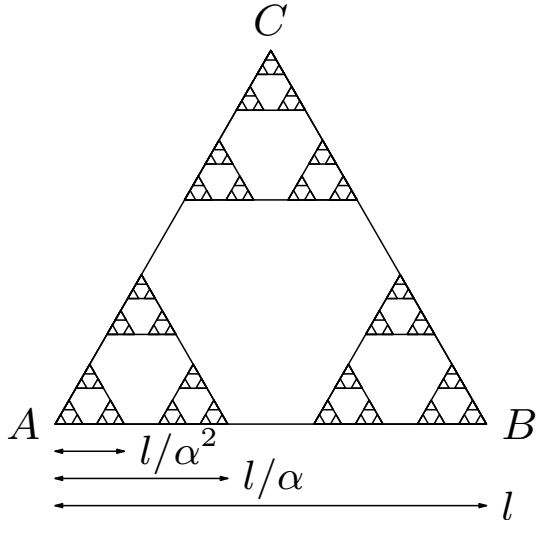
between the force exerted by the plane on the block and the inclined plane itself?

Marek found the block to be lonely on an inclined plane

Problem 35 ... long power line

8 point

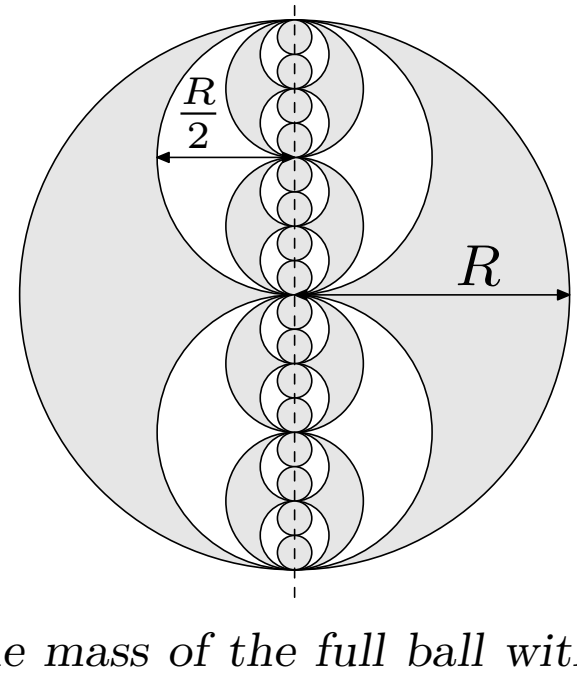
Consider a wire with constant electrical resistance $r_0 = 1\,\Omega$ per unit length l. Let us construct a resistive network from this wire as shown in the diagram. We start from a large equilateral triangle with side length l. Let us connect the points on the sides of this triangle at a distance l/α from the nearest vertex with a wire (where $\alpha \geq 2$), making an α times smaller triangle at each vertex of the large triangle. We repeat the same procedure for these smaller triangles and so on indefinitely (each step produces triangles α times smaller than in the previous step). Finally, we connect the source terminals to vertices A and B. What will the resistance of this network be if $\alpha = 4$?



Radka couldn't find the end of the extension cord

Problem 36 ... a ball for Jáchym

6 point



We will create a special ball for Jáchym in successive steps. We begin with a homogenous solid ball of radius R and material density ρ, and choose one major axis passing through the center of the ball. We remove two smaller balls of radii $R/2$ from the large ball, whose centers lie on the major axis and are at a distance $R/2$ from the center of the large ball so that the resulting holes are adjacent to each other. Then we add 4 balls inside with radii $R/4$. Their centers lie on the major axis, and we place the balls next to each other. We continue in the same way. That is, we make two spherical holes inside each of the small balls, and then put half-radius balls in them again. We continue this way to infinity. What is the ratio of the total mass of the ball for Jáchym and the mass of the full ball with the radius R and the same density ρ? *Karel wanted Jáchym to get his special ball, too*

Problem 37 ... polarization as far as it goes

6 point

Karel decided to put five linear polarizers behind each other. He chose the angle between th planes of the first and second polarizer to be α, the next $\alpha - \pi/2$, then α again, and so on What is the maximum intensity of light that Karel can obtain in this way if he shines on th first polarizer with unpolarized light of intensity I_0? Express as the ratio $I_{\max}/I_0$.

Petr was thinking about stacking polarizing filters

Problem 38 ... dog walking

7 point

Jarda is walking his dog. They are walking down a long straight street. Jarda's distance fro
he garden fences is 5.0 m *while his dog is tied to a* 4.0 m *long leash. Already from afar, the do*
nows about his friend Žeryk, who is lying contently in a kennel by the fence of one of the garden
nd is looking forward to a barking encounter. For this reason, he always tries to be as clos
o Žeryk as possible on his leash. What is the highest velocity Jarda's dog will move at? Jard
valks with a constant velocity $v = 1.0\,\mathrm{m{\cdot}s^{-1}}$.

Jarda invents problem assignments even while walking his dogs

Problem 39 ... ordinary estimate of mean air height

6 point

Consider a hollow cylinder with base area $S = 327\,\mathrm{cm}^2$, which is closed at the bottom an infinitely tall at the top. The whole cylinder is placed in a homogeneous field with a gravitationa acceleration g. Let us fill it with diatomic nitrogen molecules with temperature $T_0 = 25\,^\circ\mathrm{C}$ s that the pressure $p_0 = 101\,\mathrm{kPa}$ is at the bottom. What is the mean height of these molecule above the base of the cylinder? Assume that nitrogen behaves like an ideal gas.

Lego neglected as much as possible

Problem 40 ... modern art

7 point

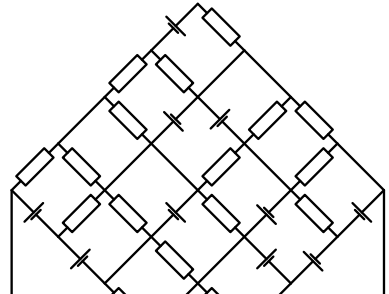
We have 2 rectangular cuboids, one with mass $M = 9.50\,\mathrm{kg}$ and the other with mass m. W
ift the second one and press it against the wall by the first one, which is standing on th
round. What is the largest possible mass m to prevent the second cuboid from falling if th
oefficient of friction between the cuboids, the cuboid and the wall, and between the cuboi
nd the ground is $f = 0,288$? Assume that the rectangular cuboid standing on the ground i
ong enough not to topple backward. *Lego really didn't know how to name this problem*

Problem 41 ... maze of resistance

6 point

All resistors (including the bulbs) in the circuit in the figure have a value of $R = 1.0\,\Omega$, capacitors have a capacitance of $C = 1.0\,\mathrm{F}$ and all batteries have a DC voltage of $U = 1.0\,\mathrm{V}$. Calculate the current through the bulb when the current is settled.

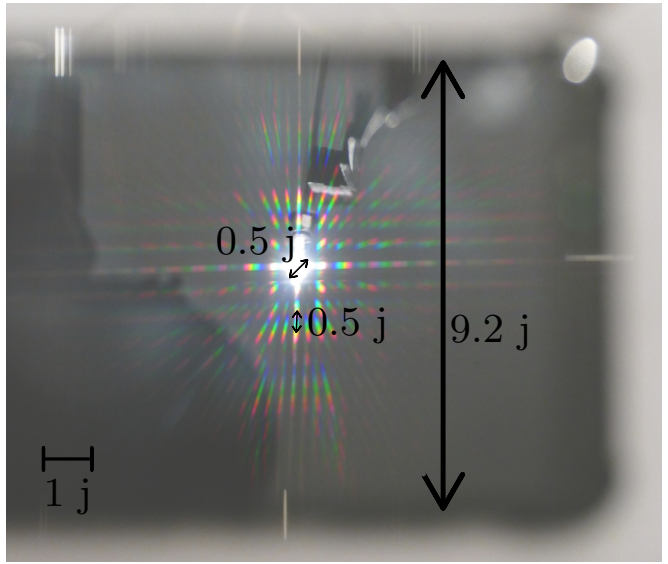
Marek J. encountered resistance while walking.



Problem 42 ... screen resolution

8 point

Jarda noticed a diffraction on his cell phone, so he tried to calculate the size of the individual pixels, which he thought were squares. He placed the cell phone 1.05 m *under a light source with a size of* 1.5 cm *and photographed the reflection on the screen with a camera whose lens was* 40 cm *above it. In the photograph, the width of the screen was* 9.2 j, *the diameter of the light was* 0.5 j, *and the distance of the yellow maxima was* 0.5 j, *where* 1 j *is the scale on the photograph. The screen of Jarda's phone has dimensions* 13 cm *and* 7.5 cm. *How many pixels are there?*



Jarda introduced an experimental problem to Physics Brawl Online

Problem 43 ... wired

7 point

Two adjacent electric transmission towers are distanced $s = 400\,\mathrm{m}$ *in a flat landscape. One c*
he wires between them is fixed on both ends at a height of $H = 20\,\mathrm{m}$ *above the ground. I*
winter, when the air temperature is $t_\mathrm{z} = -10\,^\circ\mathrm{C}$*, the lowest point of the wire is due to saggin*
t $h_\mathrm{z} = 19\,\mathrm{m}$ *above ground. At what height will this point be in summer when the temperatur*
s $t_\mathrm{l} = 20\,^\circ\mathrm{C}$*? The coefficient of thermal expansion of the steel from which the wire is mad*
s $\alpha = 13 \cdot 10^{-6}\,\mathrm{K}^{-1}$*. For simplicity, neglect the elongation of the wire due to its weight.*

Matej Rz was outside

Problem 44 ... electromagnet

8 point

Consider a long thin bar made from a material with high relative permeability $\mu_r = 7\,000$ *A wire is wrapped around the rod to obtain a solenoid with* $N = 1\,000$ *turns. The solenoid i then bent into a circle so that the opposite ends of the bar almost touch, but a narrow gap o width* $s = 1.00\,\mathrm{mm}$ *is left between them. This gap is small compared to both the radius of th circle* $r = 20.0\,\mathrm{cm}$ *and the radius of the bar. If a current* $I = 500\,\mathrm{mA}$ *flows through the wire determine the magnitude of the magnetic induction in the gap.*

Jindra stole the problem from a textbook

Problem 45 ... ladder

7 point

A ladder leaning against a wall has a length of $l = 5\,\mathrm{m}$ *and a mass of* $m = 10\,\mathrm{kg}$*, with it enter of gravity located exactly at its midpoint. When Jindra stands on the ladder, his cente f gravity is at a distance of* $h = 4\,\mathrm{m}$ *along the ladder from the ground, and a distance c* $= 30\,\mathrm{cm}$ *perpendicularly to it. Jindra's mass is* $M = 70\,\mathrm{kg}$*. The coefficient of kinetic frictio etween the ladder and the floor is* $f_1 = 0.4$ *and between the ladder and the wall is* $f_2 = 0.6$ *Determine the maximum angle between the ladder and the vertical wall, before Jindra start alling.*

Jindra painted the walls of a room

Problem 46 ... mushroom hunting

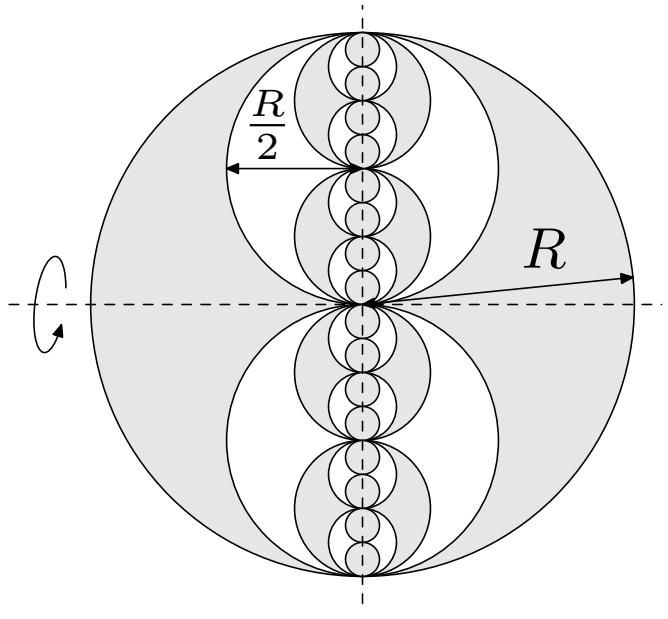
7 point

Jarda went to an oak forest to collect parasol mushrooms. The wind was blowing quite strongly o acorns from trees were falling with a frequency per unit area of $N = 1.0 \cdot 10^{-4}\,\mathrm{m}^{-2}\cdot\mathrm{s}^{-1}$ *The area that Jarda normally occupies is* $A = 1100\,\mathrm{cm}^2$*. What is the probability that an acor will hit him if he plans on picking mushrooms in the forest for* $t = 35\,\mathrm{min}$*?*

Jarda was hit by an acorn three times in a rou

Problem 47 ... a ball for Jáchym reloaded

7 point



In successive steps, we will make a special ball for Jáchym. We start with a homogenous solid ball of radius R and material density ρ. We choose one major axis passing through the center of the ball. Next, we extract two balls from the large ball that have radii $R/2$, whose centers lie on the major axis and are at distances $R/2$ from the center, such that the resulting holes are closely adjacent to each other. Then we add 4 balls of radii $R/4$ inside. Their centers again lie on the main axis and the balls are placed next to each other. We repeat this process of making two spherical holes in each of these smaller balls and then putting the half balls into these again ad infinitum. What is the moment of inertia of Jáchym's ball with respect to the axis perpendicular to the main axis? Express it as a ratio to the moment of inertia of a full ball of radius R and equal density ρ.

Karel (even Vojta and the other Vojta) thinks that Jáchym deserves the ball not once, but twice

Problem 48 ... up and down

8 point

Ping pong is played with balls of diameter $d = 4.0\,\mathrm{cm}$ *and mass* $m = 2.7\,\mathrm{g}$. *After losing a match Narda took one of these balls and out of anger fired it vertically in the air. The ball reache*

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[3] This stems trivially from the fact that the moment of inertia is the integral of the square of the distanc from the axis of rotation over the total mass of the body.

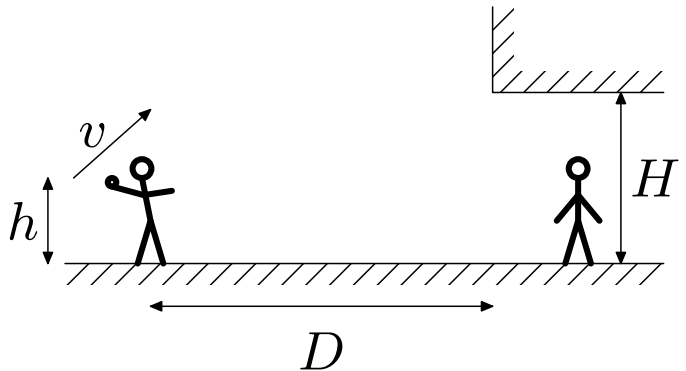
[4] More precisely, it is the integral of the square of the distance from the axis of rotation over the mass.

he height $H = 8.5\,\mathrm{m}$. Determine the ratio of the time the ball was going up and the time it wa
alling down. *Jarda tried to hit the window in his dorm with a ping-pong ball, but he failed*

Problem 49 ... revenge for the machines

7 point

Jarda got really angry at Viktor for his noisy vacuum cleaners and started chasing him around the dormitory. Viktor is now hiding in a passage under the dorms. The ceiling is $H = 5.00\,\mathrm{m}$ high there. Jarda is standing at a distance $D = 12.0\,\mathrm{m}$ from the beginning of the passage when he throws an object at Viktor with velocity $v = 13.5\,\mathrm{m\cdot s^{-1}}$ from a height of $h = 2.10\,\mathrm{m}$. How far from the beginning of the passage must Viktor stand to avoid being hit by Jarda?



Viktor created an inspiring environment in his room..

Problem 50 ... water in a bowl

7 point

We pour 650 ml *of water into a bowl shaped like a perfect homogenous hemisphere of ra*
lius 12 cm *and mass* 610 g. *Then we start tilting the bowl. What work we must do to get th*
vater to start pouring out? Assume that the water is an ideal liquid, and neglect the thicknes
f the walls. *Jarda was rinsing a bowl which had had cucumber salad in it*

Problem 51 ... where to hide?

9 point

t is the year 2311 and aliens are about to attack Earth. They arrived in their spaceship an topped at an altitude of $h = 12\,200\,\mathrm{km}$ *above the South Pole. Fortunately, the Earthlings ha lready developed a defensive missile system and destroyed the attackers' ship. It fell apart int housands of tiny fragments, which flew in all directions at speeds up to a maximum of* $v =$ $= 300\,\mathrm{m{\cdot}s^{-1}}$*. What percentage of the Earth's surface do the fragments hit? The ship was mad f refractory material, so the debris does not burn in the atmosphere.*

Jarda saw Anubis' fleet destroyed in Stargate

Problem 52 ... reading against the light

9 point

magine a large plane with an albedo (a measure of reflectivity) of 0.69*. At a height* $h = 1.0\,\mathrm{m}$ *bove the plane, there is a point source of light illuminating it. Precisely below the source, w lace a black disc with a radius* h *at a height* x*, parallel to the plane. Determine the value of* x *or which the center of the bottom face of the disc receives the greatest luminous flux. Assum hat the reflected light is scattered uniformly in all directions and the plane has a Lambertia urface. Matěj wanted to read while lying down, but the only light source was the chandelier*

Problem 53 ... rattlesnake 11

8 point

Semir and his wingman are chasing a criminal in a police car. The perpetrator drove off th road onto a horizontal forest path, thinking he could outrun the highway cops. However, he di not notice that the road ended with a cliff, $H = 20\,\mathrm{m}$ deep. The criminal is approaching th cliff with a velocity $v = 144\,\mathrm{km{\cdot}h^{-1}}$. At what angle does the front bumper hit the ground? Th parameters of the perpetrator's car are: mass $m = 1.32\,\mathrm{t}$, wheelbase $D = 2.46\,\mathrm{m}$, distance fro the front axle to bumper $d_\mathrm{n} = 0.71\,\mathrm{m}$, the height of the center of mass of the car above th ground $h_\mathrm{t} = 42\,\mathrm{cm}$, the horizontal distance of the center of mass from the front axle $d_\mathrm{t} = 80\,\mathrm{cm}$ the height of the front bumper above the ground $h_\mathrm{n} = 15\,\mathrm{cm}$, the moment of inertia of the ca with respect to the center of mass $I = 1.67 \cdot 10^3\,\mathrm{kg{\cdot}m^2}$. The car uses front wheel drive. Assum that the rear wheels roll without resistance.

Jindra watches German crime series

Problem 54 ... spring with friction

8 point

Consider a massless spring with stiffness $k = 9.0\,\mathrm{N{\cdot}m^{-1}}$. *We attach a weight with mass* $m =$ $= 110\,\mathrm{g}$ *to its free end. The other end of the spring is anchored in a wall. The spring is horizonta nd the weight slides on the floor with friction. The coefficient of friction is* $f = 0.35$, *and th est length of the spring is* $l_0 = 30\,\mathrm{cm}$. *By pulling on the weight, the spring is stretched b* $_0 = 20\,\mathrm{cm}$ *and then released with zero initial speed. When the weight comes to a complet top, what will the length of the spring be?*

Jindra was playing with a slinky

Problem 55 ... central spring

9 point

Consider a spring attached at a point $r = 1\,321\,\mathrm{km}$ *away from the center of a planet wit* *mass* $M = 4.11 \cdot 10^{22}\,\mathrm{kg}$ *and radius* $R = 1\,220\,\mathrm{km}$. *The spring has a free length* $x_0 = 89\,\mathrm{km}$ *an* *homogeneous linear density* $\lambda = 8.7\,\mathrm{kg{\cdot}m^{-1}}$. *What is the stiffness which the spring must hav* *f we require it to exactly reach the surface of the planet? Do not consider any rotation of th* *ystem.*

Jáchym was sorr *hat he had forgotten about a problem with a non-massless spring, so he decided to correct i*

Problem 56 ... foggy glass

9 point

Consider horizontal glass with thickness $d_g = 16.3\,\mu m$ *and refractive index* $n_g = 1.68$. *We spi*
water with refractive index $n_w = 1.33$ *on the glass in such a way that it forms a layer wit*
uniform thickness $d_w = 15.5\,\mu m$. *From the bottom, we shine a vertical beam of light wit*
wavelength $\lambda = 0.590\,\mu m$ *on the glass. What is the ratio of the intensity of light which passe*
through in this situation to the intensity of light which would pass through if there was n
water? Assume that light is only reflected or transmitted by the materials, not absorbed.

Jarda's window got foggy

Problem 57 ... Matěj's black warm box

7 point

Determine the number of photons in a sealed box of exact volume $1\,\ell$*, which is in therma*
equilibrium in a room with a temperature of $25.0\,\mathrm{K}$*.*

Karel was wondering what the head of Physics Brawl Online does when out of the public eye

Problem M.1 ... fast and even faster for the first time

3 point

Two cars travel the same distance s. Both are moving with constant acceleration. The firs moves with acceleration a_1, and the second with acceleration $a_2 = 1.25\,a_1$. The initial velocit of both cars is zero. How fast will the second car travel the required distance? Enter the resul into the system as a ratio of the second car's time to the first car's time.

Karel varied the problems

Problem M.2 ... fast and even faster for the second time

3 point

We have two cars, and we want them to travel the distance s in the same time t. During thei
ourney, they will both have constant acceleration in the direction of travel. The first car wi
ravel with an acceleration of a_1, whereas the second car will have an acceleration of
$_2 = 1.250\, a_1$. The second car starts its movement from the rest. With what initial velocity v
must the first car start? Give the result as a multiple of $a_1 t$ (enter only the number by whic
ou multiply these quantities). *Karel varied the problems for the second time*

Problem M.3 ... fast and even faster for the third time

3 point

Two cars have to travel the same distance s. Both will be moving with the constant jerk, whic
s a change in acceleration over time (analogous to acceleration being a change in velocity). Th
irst car will travel with jerk g_1 and the second with jerk $g_2 = 1.25\, g_1$. The initial velocities an
ccelerations of both cars are zero. How fast will the second car travel the required distance
Enter the result into the system as the ratio of the second car's time to the first.

Karel varied the problems for the third time

Problem M.4 ... fast and even faster for the fourth time

3 point

Two cars have to travel the same distance s. Both will move with constant power. The powe
of the first car will be P_1, and the power of the second car will be $P_2 = 1.25\,P_1$. The initia
velocity of both cars is zero, and they have the same mass, including the load. How fast wi
he second car travel the required distance? Enter the result into the system as the ratio of th
econd car's time to the first. *Karel varied the problems for the fourth time*

Problem E.1 ... light circle

4 point

A thin convex lens is placed at a distance of 7.0 cm *from the screen. We then turn on a distan light source located on its optical axis. An illuminated circle appears on the screen behin the lens. After moving the lens to a distance of* 9.0 cm *from the screen, we notice that the circl on the screen now has the same radius as before. What is the focal length of the lens?*

Jarda saw the opposite of the eclipse

Problem E.2 ... planparallel lens

3 point

A thin convex lens with radii of curvature $R_1 = 20\,\mathrm{cm}$ and $R_2 = 35\,\mathrm{cm}$ is cut perpendicular t
ts optical axis. The two halves are placed with the convex sides close to one another (the fla
urfaces are facing away from each other). What is the ratio between the old and the new foca
engths of this arrangement of glasses? The refractive index of the glass is $n = 1.5$.

According to Jarda, shape does not matter

Problem E.3 ... two magnifiers

4 point

Jarda found two magnifying glasses at home and started playing with them. When he pu hem close together, he discovered they had a focal length of $F = 7\,\mathrm{cm}$. *He then placed th enses at a suitable distance apart, stood far behind them, and looked at a distant tree. He sa t sharply and its magnitude was* 1.5*-times larger than without the lenses. What is the differenc*

n the focal lengths of the two lenses? Subtract the smaller value from the larger. Think o
he magnifying glass as a thin lens.

Jarda lives so high that he can't really see the ground from the windou

Problem E.4 ... recording ants

8 point

We place two lenses and a screen on an optical axis. One lens, with focal length $f_1 = 3.6\,\mathrm{cm}$ *s located* $d = 8.3\,\mathrm{cm}$ *in front of the screen. The second lens with focal length* $f_2 = -1.6\,\mathrm{cm}$ *i ocated between the first lens and the screen. An ant is initially located* $a_0 = 9.7\,\mathrm{cm}$ *in fron f the first lens and is walking along the optical axis towards the lenses and the screen wit elocity* $v = 0.6\,\mathrm{cm{\cdot}s^{-1}}$*. The image of the ant on the screen is initially in focus. What shoul e the velocity of the second lens if we want the image to stay in focus for the next momen luring the walk? The sign of this velocity should be positive if the lens is moving in the sam irection as the ant.*

Jarda likes to take photos of insects

Problem X.1 ... we are cleaning up the garden

4 point

Autumn has come, and the leaves in our garden have started to fall. After raking them, we need to take them to a nearby pile, so we scoop them on a garden wheelbarrow and hea towards the pile at a constant speed $v = 1.2\,\mathrm{m\cdot s^{-1}}$*. We are pushing the wheelbarrow in front o us on a pavement made of interlocking paving stones, so there is always a small hole betwee the* 15 cm *tiles, where the speed of the wheelbarrow is reduced by* 10 %*. Determine the constan horizontal component of the force with which we need to push the wheelbarrow forward to mak its average speed also* v*. The wheelbarrow together with the leaves has a mass of* 19 kg*.*

Many leaves have fallen on Jarda's greenhouse

Problem X.2 ... watering the garden

4 point

Jarda planted six fruit bushes next to a garden pond with a surface area of $A = 8\,\mathrm{m}^2$*. Thes tand in a row, the first being* $2.3\,\mathrm{m}$ *from the pond and every other* $1.0\,\mathrm{m}$ *further than the pre ious one. Jarda always waters them using his ten-liter watering can that he refills at the pond The total time it takes him to refill and pour one jug is* $45\,\mathrm{s}$*. Jarda walks at a speed of* $v = 1.2\,\mathrm{m{\cdot}s^{-1}}$*. In the meantime, the garden hose delivers water to the pond with a flow rat of* $Q = 0.9\,\mathrm{l{\cdot}s^{-1}}$*. After watering the last plant, he went to see how much the level of the pond ha isen since he had started filling the first watering can. How much did he measure? If the leve has dropped, submit a negative value.*

Jarda found currants on sale

Problem X.3 ... winding the cable

6 point

Jarda mows his garden with an electric mower. Once he has finished, he has to wind the entir cable of $L = 20\,\mathrm{m}$ *onto the cable reel. It is wound onto a cylinder with a diameter* $D = 40\,\mathrm{cm}$ *whose axis of symmetry is horizontal. The cable attachment on the reel is close to the ground The coefficient of friction between the ground and the cable is* $f = 0.7$*, and the length densit of the cable is* $\lambda = 220\,\mathrm{g{\cdot}m^{-1}}$*. Determine the amount of work that Jarda will do when winding Jarda was laying the cable*

Problem X.4 ... swinging cucumber

6 point

A cucumber in Jarda's greenhouse has the shape of an elongated ellipsoid with semi-axes $a = 3\,\mathrm{cm}$ *and* $b = 10\,\mathrm{cm}$ *and hangs in the middle of a stem of length* $2l = 90\,\mathrm{cm}$. *The ends of the stem are distanced* $L = 51\,\mathrm{cm}$ *from each other and the weight of the cucumber is* $m = 450\,\mathrm{g}$. *Determine the period of the small oscillations around the equilibrium position when the deflection is perpendicular to the plane of the suspension.*

Jarda likes cucumbers.

